Name: ______________________

Worksheet

Problem solving with numerical summaries

Task A: Critically evaluating statistical conclusions

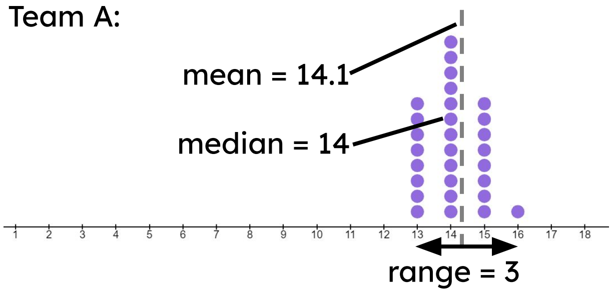
1) The table shows a statistical summary of data from The Met Office from 1941-2022. The data shows the amount of rainfall (mm) per month in Durham and Lerwick.

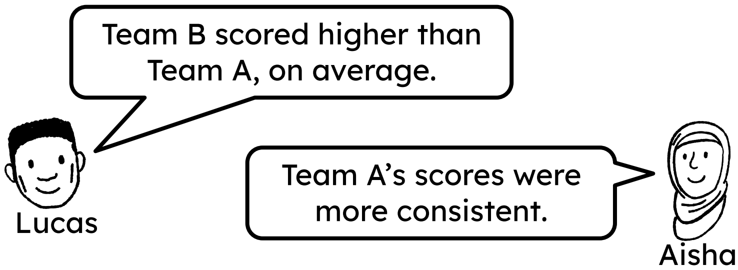
Make two comparisons between the amounts of rainfall in Durham and Lerwick.

monthly rainfall (mm)	Durham	Lerwick
mean	53.6	98.3
median	48.4	92
range	194.2	301.7

2) Two classes practised throwing basketballs into a hoop. Each person had 20 attempts. The dot plots show the number of points each person scored. Lucas and Aisha used the data to make conclusions.

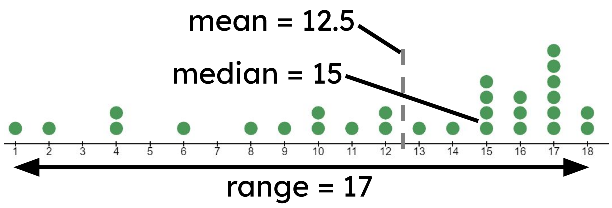
Team A:





a) Give evidence to support Lucas's conclusion.

Team B:



b) Give evidence to counter Lucas's conclusion.

c) Give evidence to support Aisha's conclusion.

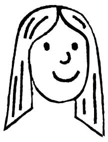
Task B: Considering context behind statistical conclusions

1) Alex has represented his school and his county in swimming races.

When I raced for my school, I got a gold medal.
But then when I raced for my county, I only got a bronze medal.
That must mean I swam slower when racing for my county.

Explain why Alex's conclusion could be wrong.

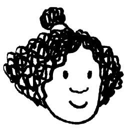
2) Laura is looking at her school reading book records.



When I was 6 years old, I read an average of 9.2 books per month.
I now read an average of 1.4 books per month.
I must be a slower reader now than I was when I was 6 years old.

Explain why Laura's conclusion could be wrong.

3) A survey found that 18% of workers in the town of Scarborough walk to work and that 7% of workers in the City of London walk to work.

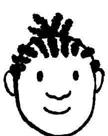


There are more people walking to work in Scarborough than in London.

Izzy

Explain why Izzy's conclusion could be wrong.

4) A survey in 1971 found that 15% of workers in Liverpool travelled to work by bus. A survey in 2023 found that 11% of workers in Liverpool travelled to work by bus.



Liverpool does not need to run as many buses as they did in 1971 because there are fewer people using them to travel to work.

Jacob

Explain why Jacob's conclusion could be wrong.

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